

05

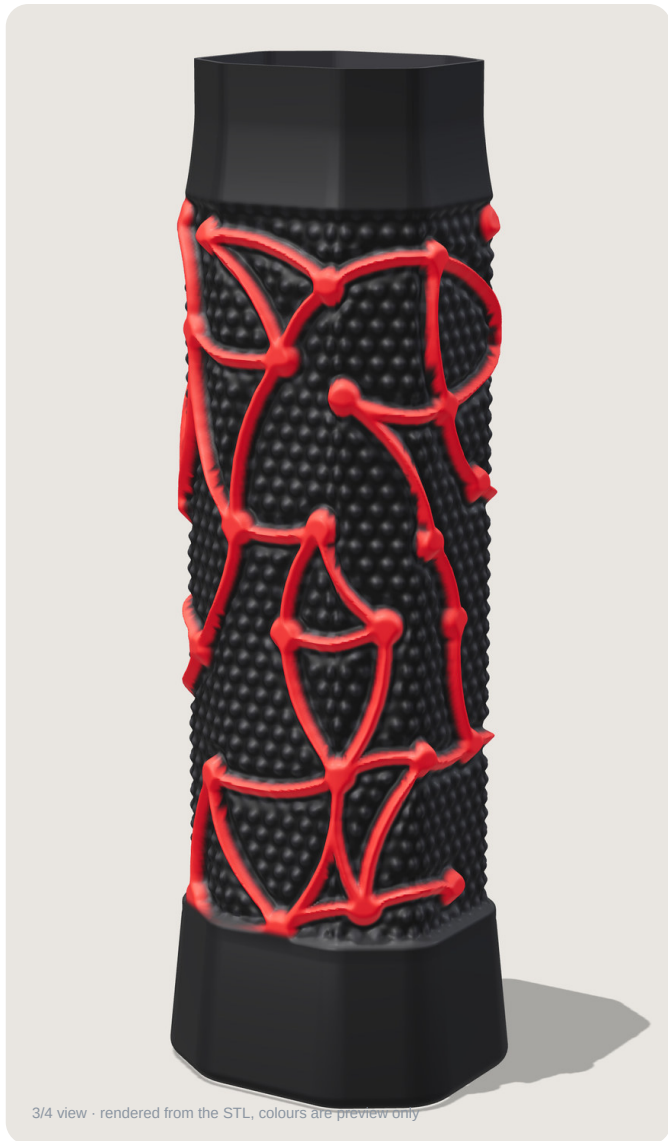
NEURO-TREAD

Bio-inspired raised neural network over a dimpled adaptive base.

46.5 g

PLA · 100 % infill · solid

✓ UNDER 50 G IN ALL 4 MATERIALS



3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	132.82 mm
Max footprint (foot)	42.63 × 36.61 mm
Bore	32.06 × 26.04 mm, 127.8 deep
Textured band	z 21.5 – 116.0 mm (94.5 tall)
Outer size in band (max)	39.1 × 33.0 mm
Peak texture height	3.50 mm above bore
Base wall / min rim	1.2 mm / 0.8 mm
Volume / surface area	37.49 cm ³ / 330 cm ²

WEIGHT · 100 % INFILL

PLA	1.24 g/cm ³		46.5 g
PETG	1.27 g/cm ³		47.6 g
TPU	1.21 g/cm ³		45.4 g
ABS	1.04 g/cm ³		39.0 g

50 g limit

COLOURWAY



3MF print palette

PRINT PROFILE

Infill / layer height	100 % / 0.20 mm
Layers	665
Orientation	upright, butt end down
Supports	none required
Filament, 1.75 mm	≈ 15.6 m (100 % flow)

DESIGN INTENT

A network of 48 neurons joined by curved axon ridges rises to the full 3.5 mm crest, standing proud of a field of small bumps. The ridges guide the fingers while the bump field provides fine grip.

Ridges and bumps are gentle slopes; the network is printed in the same material as the base.

VERIFIED AGAINST THE REFERENCE

Bore vs reference	identical
Outer envelope vs reference	+0.001 mm at most
Rim vs reference	+0.40 mm (deliberate, ≥ 0.8 mm wall)
Bodies / enclosed voids	1 / none
Material under 0.4 mm wide	0.000 % of volume
Longest bridge	none

KEY FEATURES

- 48 somas (8 × 6) joined by curved axons about 1.5 mm wide
- Bump field on a 2.66 mm hex pitch
- Network and somas reach the full 3.5 mm crest

FILES

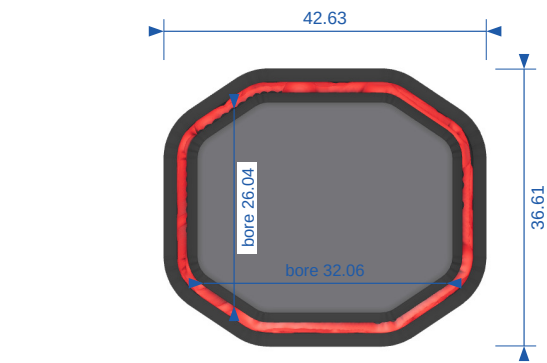
stl/05_neuro_tread.stl	19.9 MB · 398,860 triangles
3mf/05_neuro_tread.3mf	4.8 MB · colour

✓ FITMENT VERIFIED

✓ SOLID · ONE CLOSED BODY

✓ PRINT-READY · 0.20 MM LAYERS

✓ 3MF VALIDATED



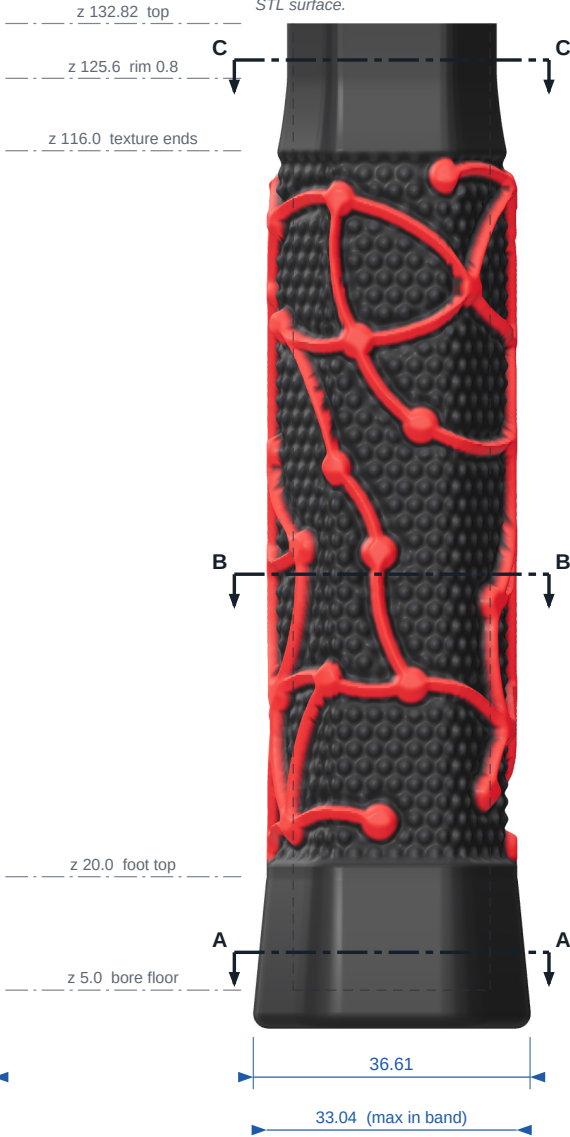
TOP VIEW (from +Z)



FRONT VIEW (from -Y)

KEY DIMENSIONS	
Overall height	132.82
Overall width × depth (foot)	42.63 × 36.61
Bore width × depth	32.06 × 26.04
Bore depth (floor at z 5.0)	127.82
Outer W × D, textured band (max)	39.06 × 33.04
Foot top (z 20)	39.16 × 33.13
Butt face (z 0)	40.56 × 34.54
Peak texture height above bore	3.50
Base wall / minimum rim wall	1.20 / 0.80

All values in mm. Dashed lines: bore (hidden edges). Renders are the actual STL surface.



RIGHT VIEW (from +X)

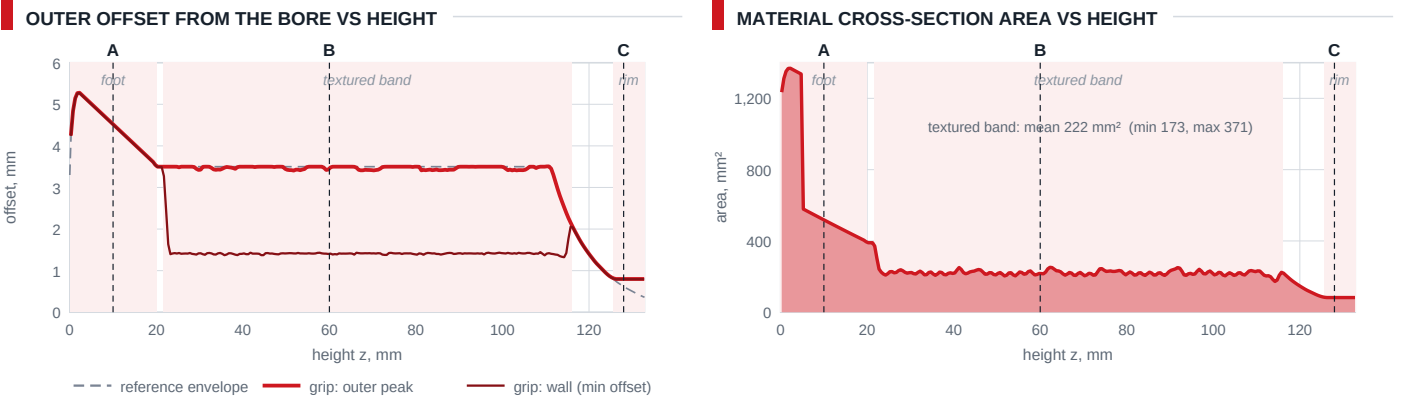
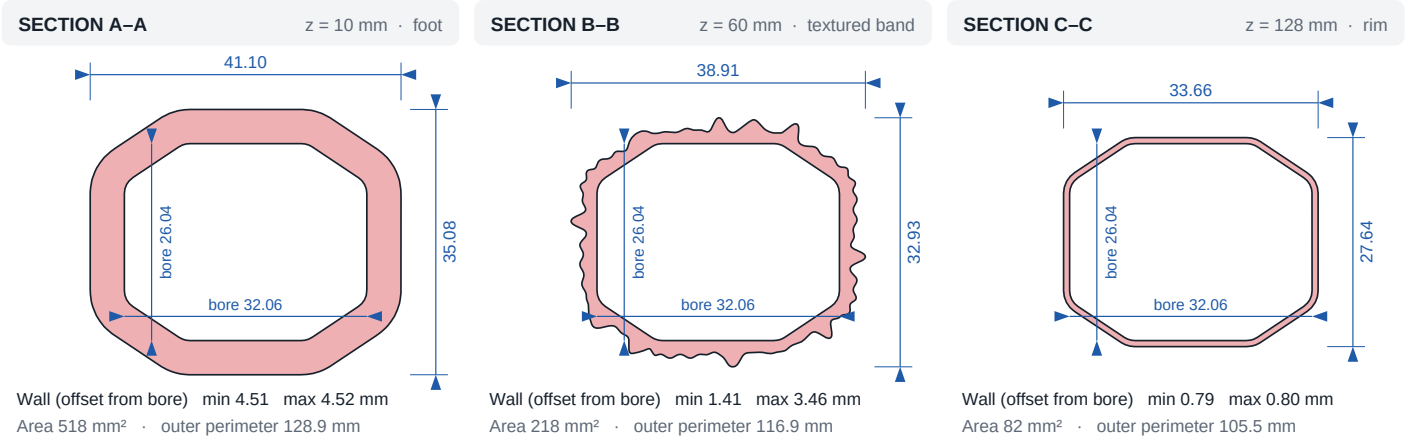
A, B, C: see sheet 3



PART NEURO-TREAD Pickleball paddle grip sleeve, closed butt, open throat	SCALE 1 : 1	UNITS millimetres	PROJECTION
	MATERIAL PLA / PETG / TPU	INFILL 100 %	LAYER HEIGHT 0.20 mm
GENERAL NOTES 1. Views are true size: print this sheet at 100 %, do not fit to page. 2. Dimensions are measured from the STL, not nominal. 3. Bore, floor and outer envelope match the reference PickleballGrip_1.3mf. 4. Rim is held at a minimum of 0.8 mm (reference tapers to 0.36 mm).	MASS (PLA) 46.5 g	VOLUME 37.49 cm³	SURFACE AREA 330 cm²
	REVISION 2	DATE 2026-09-30	SHEET 2 of 4

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.



OVERALL & BORE

Overall length (Z)	132.82 mm
Overall width (X)	42.63 mm
Overall depth (Y)	36.61 mm
Bore width × depth	32.06 × 26.04 mm
Bore area / perimeter	751.0 mm ² / 100.5 mm
Bore depth (floor at z 5.0)	127.82 mm
Butt face (z 0)	40.56 × 34.54 mm
Outer W × D in band (max)	39.06 × 33.04 mm
Circumference in band	114.1 – 119.7 mm
Volume / surface area	37.49 cm ³ / 329.7 cm ²

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	5.0
Flared foot	0.0	20.0	20.0
Smooth collar	20.0	21.5	1.5
Textured band	21.5	116.0	94.5
Smooth neck (taper)	116.0	125.6	9.6
Rim (0.8 mm wall)	125.6	132.82	7.3
Base wall (design / measured min)			1.20 / 1.37 mm
Rim wall (design / measured min)			0.80 / 0.79 mm
Peak texture height above bore			3.50 mm

OUTER DIMENSIONS BY HEIGHT

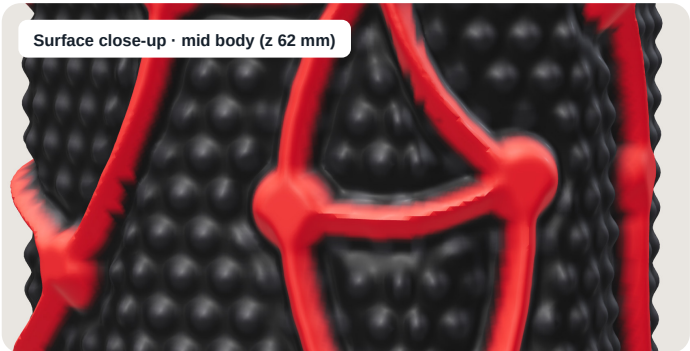
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM ²	WALL MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	958	5.00	5.01
10.0	41.10	35.08	128.9	518	4.51	4.52
20.0	39.11	33.09	122.6	393	3.52	3.52
30.0	37.77	31.75	116.8	222	1.42	3.43
45.0	38.99	32.88	117.1	234	1.40	3.50
60.0	38.91	32.93	116.9	217	1.41	3.46
75.0	38.94	32.96	119.3	232	1.42	3.50
90.0	38.99	32.86	117.2	231	1.42	3.50
105.0	38.74	32.79	116.7	208	1.41	3.42
116.0	36.24	30.19	113.5	221	2.05	2.10
125.0	33.77	27.75	105.8	88	0.85	0.85
132.0	33.66	27.64	105.5	82	0.79	0.80

Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.



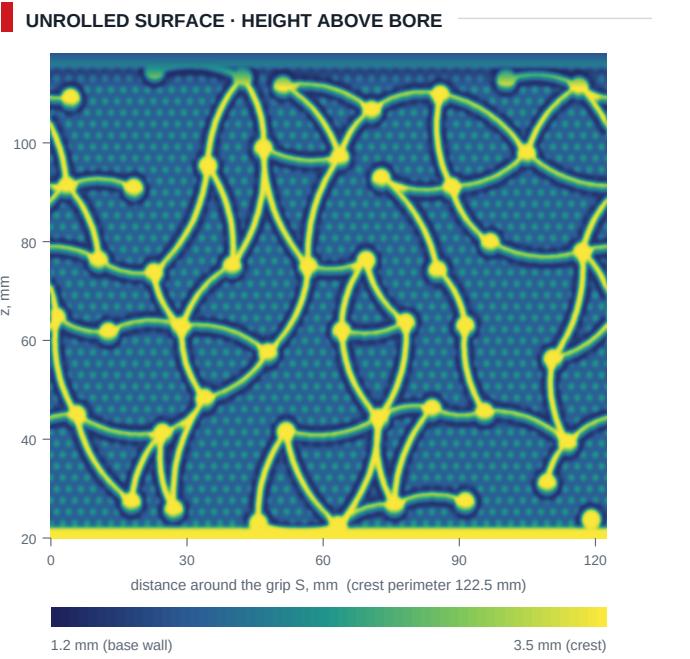
Four views, 90° apart · rendered from the STL, colours are preview only



Surface close-up · mid body (z 62 mm)



Neck, rim and bore · from above (z 119 mm)



DESIGN PARAMETERS	
Neurons (somas)	48 (8 × 6)
Axon ridge width	about 1.5 mm
Soma diameter	about 4.2 mm
Network height	3.50 mm above bore
Bump pitch	2.66 mm (hex)
Bump field height	1.4 – 2.4 mm above bore
Ridge / bump colours	red / black
PRINT & VERIFICATION	
Result: fitment / print-ready	PASS / PASS
Bodies · genus	1 · 0
Bore vs reference	identical
Envelope over reference	+0.001 mm
Material under 0.4 mm wide	0.000 %
Longest bridge	none